The Chain Rule

Section 1: Basic Chain Rule & Composite Functions

1 Differentiate $y = (3x^2 - 5x + 2)^4$.

a. $4(3x^2 - 5x + 2)^3(6x - 5)$

b. $4(6x - 5)^3$

c. $12x(3x^2 - 5x + 2)^3$

d. $4(3x^2 - 5x + 2)^3$

2 If $h(x) = f(g(x))$ with $g(3) = 5, g'(3) = -2,$ and $f'(5) = 7$, what is $h'(3)$?

a. -14

b. 10

c. -10

d. 35

3 Find $\frac{d}{dx}[\cos(4x^3)]$.

a. $-12x^2\sin(4x^3)$

b. $-4\sin(4x^3)$

c. $12x^2\sin(4x^3)$

d. $-\sin(12x^2)$

Section 2: Exponential & Logarithmic Chain Rule

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4 Find the derivative of $f(x) = e^{5x^2 - 2x}$.

a. $(10x - 2)e^{5x^2 - 2x}$

b. $e^{10x - 2}$

c. $10xe^{5x^2 - 2x}$

d. $(5x^2 - 2x)e^{5x^2 - 2x}$

5 Differentiate $y = \ln(x^3 + 4)$.

a. $\frac{3x^2}{x^3 + 4}$

b. $\frac{1}{x^3 + 4}$

c. $\frac{3x^2}{x}$

d. $\frac{1}{3x^2}$

6 Find $\frac{d}{dx}[\ln(\cos x)]$.

a. $-\tan x$

b. $\tan x$

c. $\cot x$

d. $-\cot x$

Section 3: Repeated Chain Rule Applications

7 Find the derivative of $y = \sin^3(2x)$.

a. $6\sin^2(2x)\cos(2x)$

b. $3\sin^2(2x)\cos(2x)$

c. $6\sin(2x)\cos(2x)$

d. $3\cos^2(2x)$

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8 Differentiate $f(x) = \sqrt{1 + \tan(3x)}$.

a. $\frac{3\sec^2(3x)}{2\sqrt{1+\tan(3x)}}$

b. $\frac{\sec^2(3x)}{2\sqrt{1+\tan(3x)}}$

c. $\frac{3\sec^2(3x)}{\sqrt{1+\tan(3x)}}$

d. $\frac{1}{2\sqrt{1+\tan(3x)}}$

9 If $g(x) = e^{\sin(4x)}$, what is $g'(0)$?

a. 4

b. 1

c. 0

d. $4e$

Section 4: Table-Based Chain Rule Problems

10 Let $h(x) = [f(x)]^3$. If $f(1) = 2$ and $f'(1) = -4$, what is $h'(1)$?

a. -48

b. -24

c. 12

d. -16

11 Let $k(x) = f(x^2)$. If $f'(4) = 5$, what is $k'(2)$?

a. 20

b. 10

c. 5

d. 40

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12 If $F(x) = f(g(x))$ and $g(1) = 3, g'(1) = 2, f(3) = 5, f'(3) = 6$, find the equation of the tangent line to F at $x = 1$.

a. $y - 5 = 12(x - 1)$

b. $y - 3 = 12(x - 1)$

c. $y - 5 = 6(x - 1)$

d. $y - 5 = 2(x - 1)$

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